Welcome to Elastic Amdavad Meetup!

Who Am I?

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Elasticsearch Logstash Kibana

Let’s Make this Interactive?

Make a promise( )

Let’s understand Who is Here?

Let’s understand Who is Here?

- AI Engineers or Backend Developers? (anyone working with Vectors?)
- SREs or DevOps or FinOps engineers here?
- Any students or learners exploring AI for the first time?

Why Am here?

Disk. RAM. CPU. Float32. BBQ.

The Evolution of High-Performance Vector Search

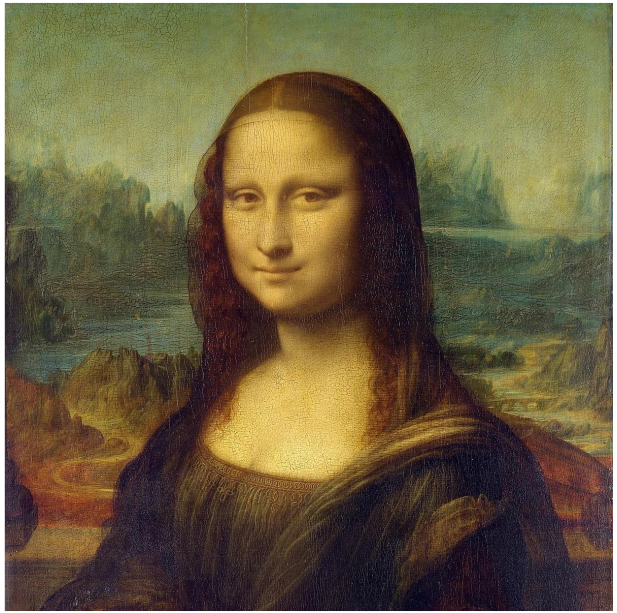
Quick Activity - Guess the Image/Person/Movie?

(If you know, don’t say it, just raise your hand)

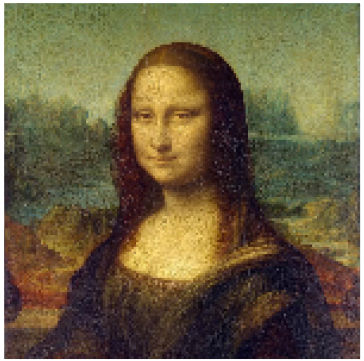
Babu Bhaiya!

1 billion vectors @ 1024 dimensions

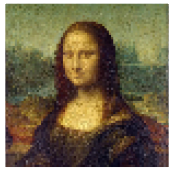
64 nodes



16 nodes



8 nodes



= 32× RAM savings (vs. float32)

2 nodes

float32 | int8 | int4 | bit | **BBQ**

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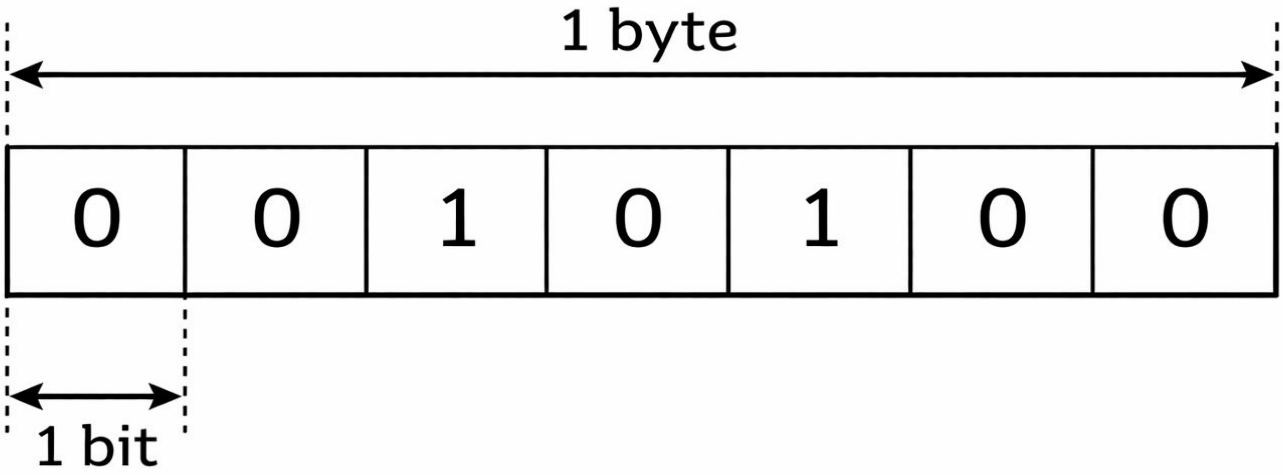
What’s the Problem?

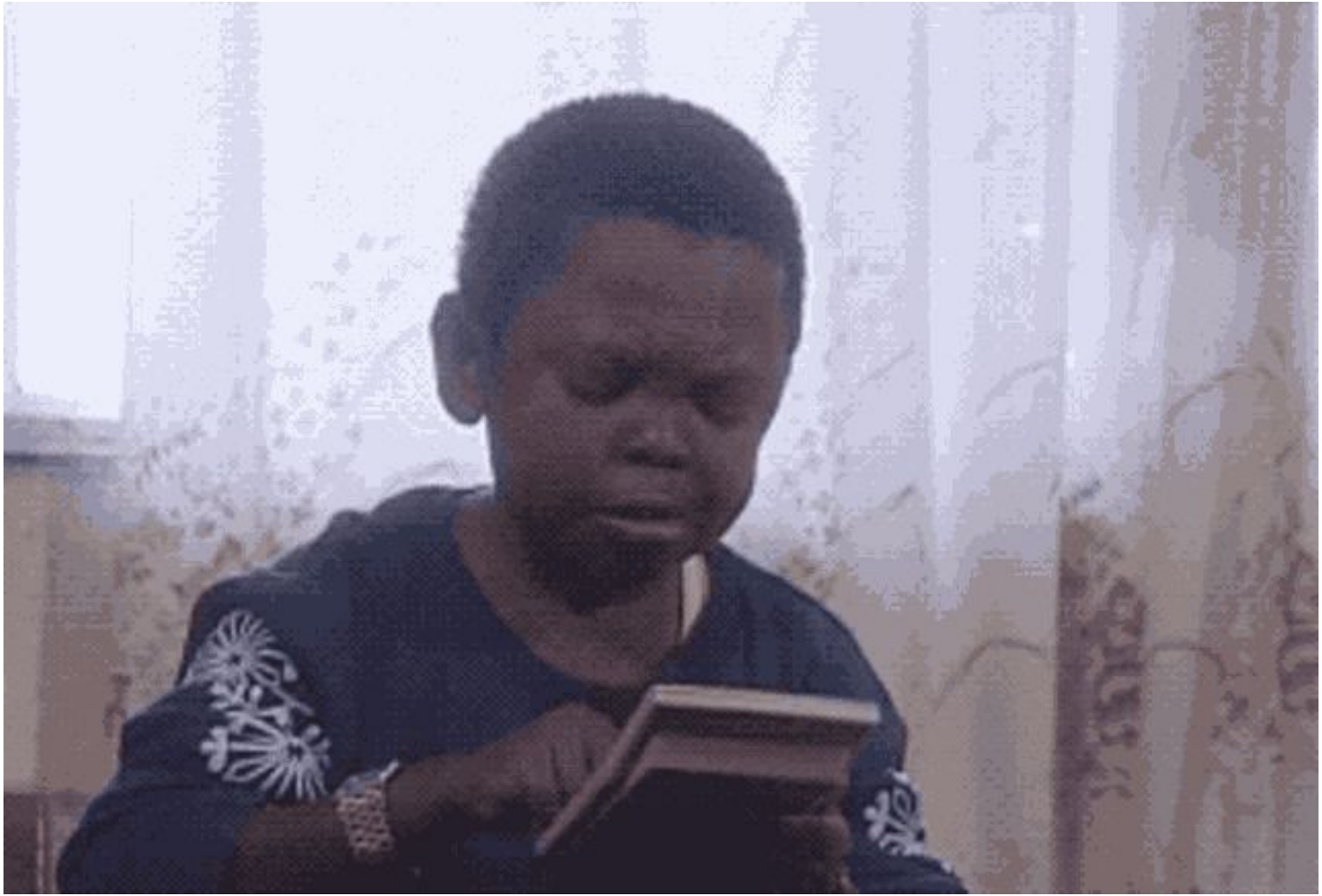
Why are we talking about it now?

(for more details, holding accurate information, you need more resources)

Let’s do some Math

Q1: How many Bits are 1-byte?





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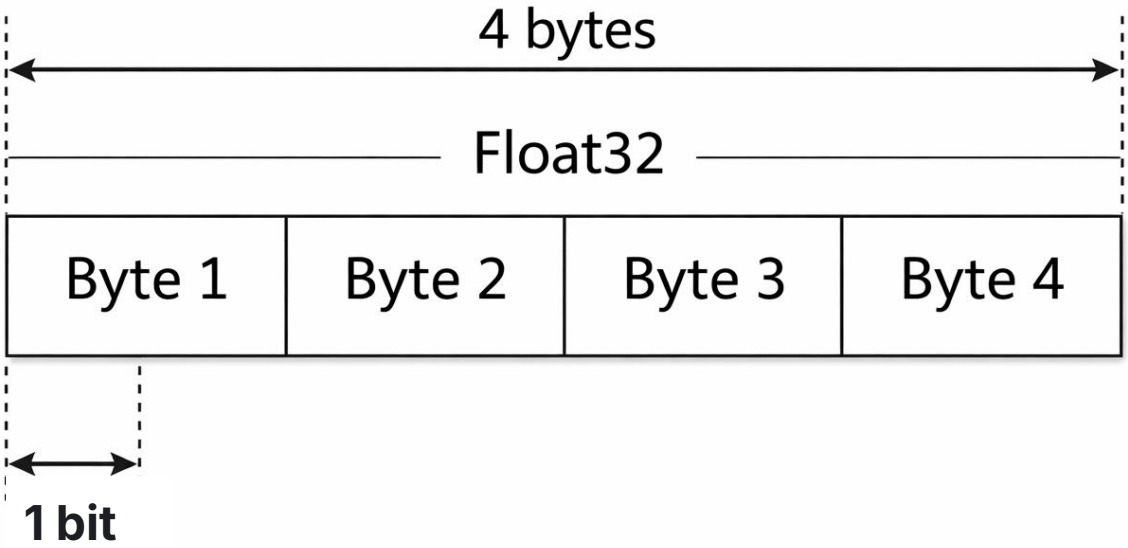
Q2: How many Bits are 1-float32 vector?

Q3: How many Bytes are 1-float32 vector?



Q4: How many Bytes does a float32 Image (100 × 100 pixels) need?

Data Traversal!

Numbers you should know

Latency Numbers Everyone Should Know

Operation	Time in ns	Time in ms (1ms = 1,000,000 ns)
L1 cache reference	1	
Branch misprediction	3	
L2 cache reference	4	
Mutex lock/unlock	17	
Main memory reference	100	
Compress 1 kB with Zippy	2,000	0.002
Read 1 MB sequentially from memory	10,000	0.010
Send 2 kB over 10 Gbps network	1,600	0.0016
SSD 4kB Random Read	20,000	0.020
Read 1 MB sequentially from SSD	1,000,000	1

Vector from memory

Vector from disk

Lovingly borrowed from:
https://static.googleusercontent.com/media/sre.google/en//static/pdf/rule-of-thumb-latency-numbers-letter.pdf

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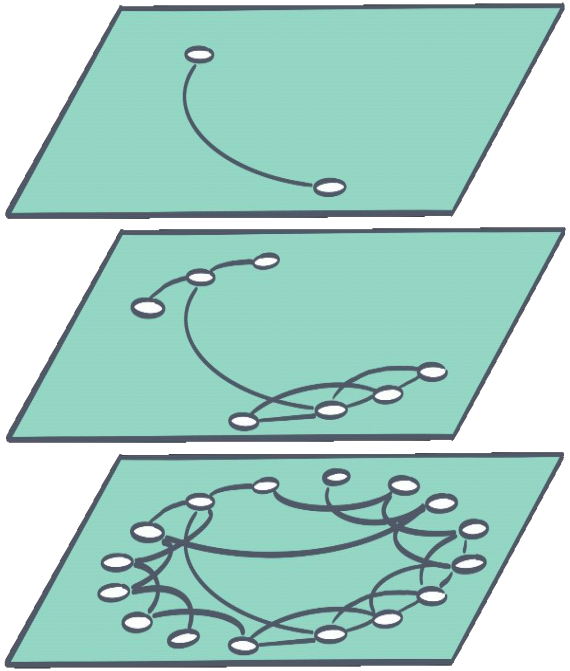
Vector Retrieval+Storing Techniques

HNSW (RAM-Resident)

diskBBQ (disk-resident vectors + RAM metadata)

HNSW retrieval

- Graph based indices (like Vamana & HNSW) are fastest
- Logarithmic search scale
- Random access of single vectors



Linear scaling is for losers

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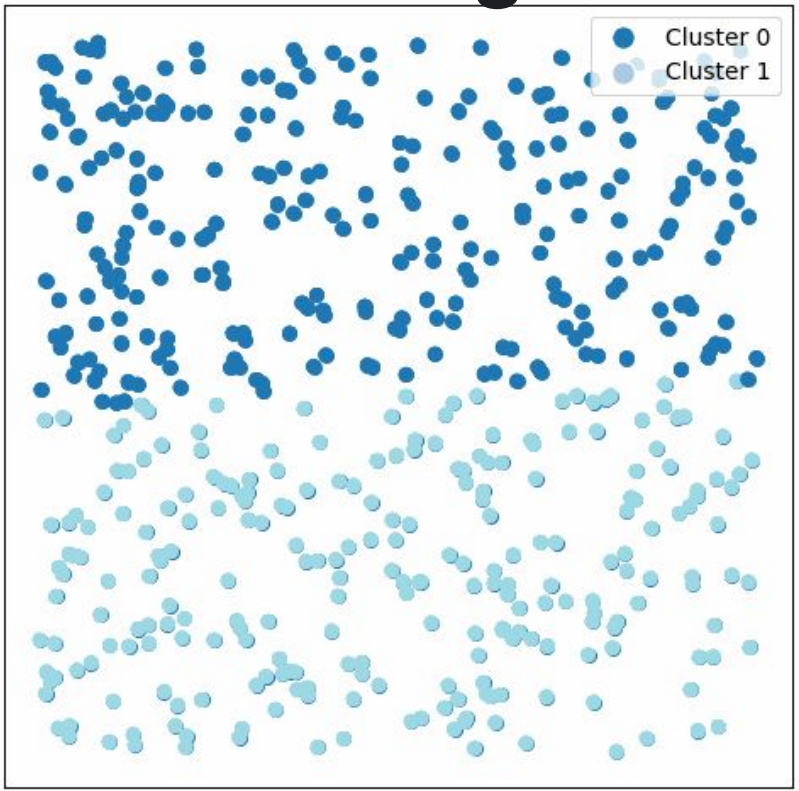
What’s the Problem now?

Did we find it?

DiskBBQ to the Rescue!

Clusterize the vectors while Indexing!



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Home

Demo Dashboard

RAM Disk Calculator

About

Light Mode

Parameters

Number of vectors

1000000 – +

Dimensions

1024

Element type

float

Quantization

none

HNSW Graph

Include HNSW graph

m (connections per node)

16

DiskBBQ

Include DiskBBQ estimate

RAM & Disk Calculator

Estimate memory and storage for HNSW + quantization using Elastic docs formulas.

Vector RAM

3.81 GB

float / none

HNSW Graph RAM

61.0 MB

m = 16

DiskBBQ Total

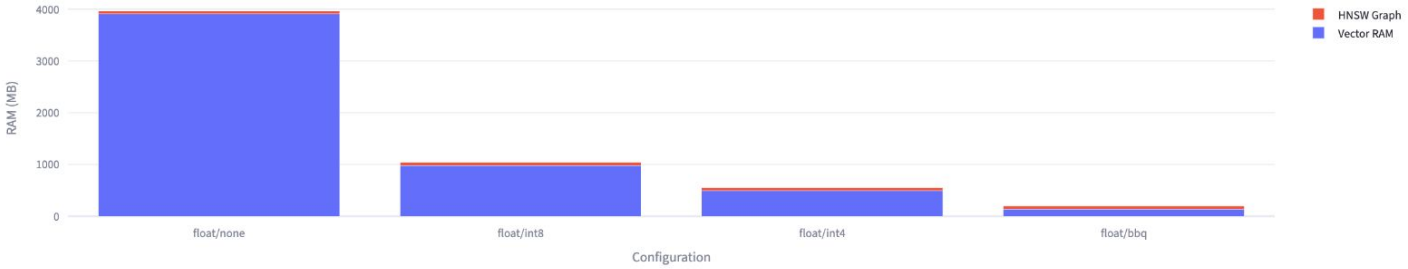
Disabled

Total RAM Estimate

3.87 GB

Vector + Graph + DiskBBQ hot

RAM Comparison Across Configurations



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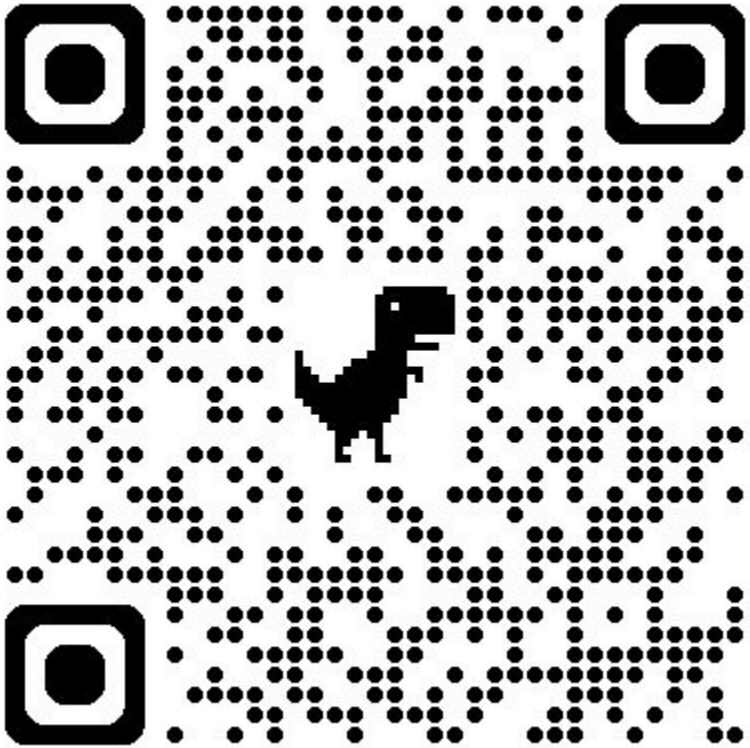


Calculator Link

If you're using HNSW, the graph must also be in memory. To estimate the required bytes, use the following formula below. The default value for the HNSW `m` parameter is `16`.

$$\begin{aligned} estimated\ bytes &= num_vectors \times 4 \times m \\ &= num_vectors \times 4 \times 16 \end{aligned}$$

The following is an example of an estimate with the HNSW indexed `element_type: float` with no quantization, `m` set to `16`, and `1,000,000` vectors of `1024` dimensions:

$$\begin{aligned} estimated\ bytes &= (1,000,000 \times 4 \times 16) + (1,000,000 \times 4 \times 1024) \\ &= 64,000,000 + 4,096,000,000 \\ &= 4,160,000,000 \\ &= 3.87GB \end{aligned}$$

If you're using DiskBBQ, a fraction of the clusters and centroids need to be in memory. When doing this estimation, it makes more sense to include both the index structure and the quantized vectors together as the structures are dependent. To estimate the total bytes, first compute the number of clusters, then compute the cost of the centroids plus the cost of the quantized vectors within the clusters to get the total estimated bytes. The default value for the number of `vectors_per_cluster` is `384`.

$$num_clusters = \frac{num_vectors}{vectors_per_cluster} = \frac{num_vectors}{384}$$

$$\begin{aligned} estimated\ centroid\ bytes &= num_clusters \times num_dimensions \times 4 \\ &+ num_clusters \times (num_dimensions + 14) \end{aligned}$$

$$estimated\ quantized\ vector\ bytes = num_vectors \times ((num_dimensions/8 + 14 + 2) \times 2)$$

Note that the required RAM is for the filesystem cache, which is separate from the Java heap.

Source: https://www.elastic.co/docs/deploy-manage/production-guidance/optimize-performance/approximate-knn-search#_ensure_data_nodes_have_enough_memory

Your data is more than just vectors

STRING
FLOAT
DENSE_VECTOR
GEO_POINT
DATE
NESTED OBJECT

```
{
  "name": "Bar Cecil",
  "star_rating": 4.8,
  "description": "The best restaurant and bar in Palm Springs, CA..",
  "description_embedding": [0.2, 0.12, 0.5, 0.22, 0.97, 0.32, 0.74, 0.49,...]
  "location": {
    "lat": 41.12, "lon": -71.34
  },
  "@timestamp": "2024-12-22T00:00:26.464Z",
  "owners_and_operators": [
    {
      "first": "John",
      "last": "Smith",
      "position": "bar tender"
    },
    {
      "first": "Alice",
      "last": "White",
      "position": "head chef"
    }
  ]
}
```

Elasticsearch Platform

INVERTED INDICES

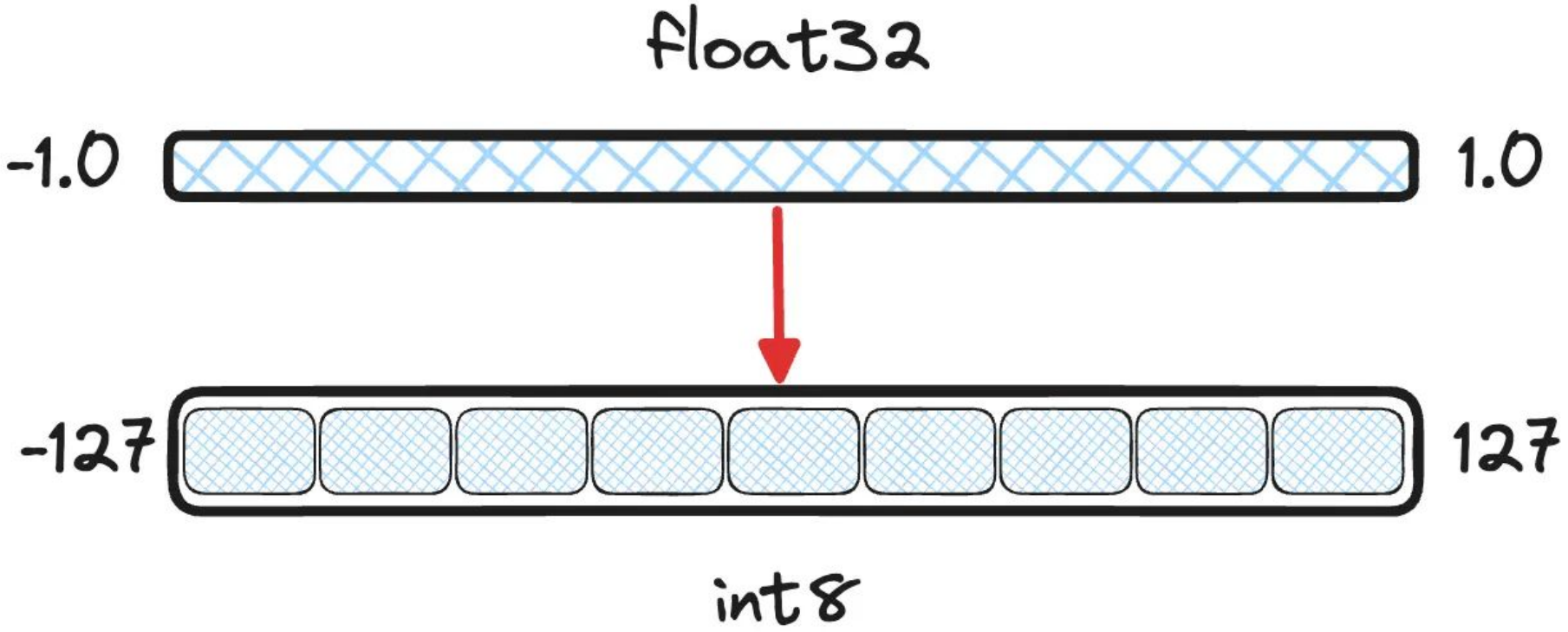
HNSW

OPTIMIZED BKD TREES

COLUMNAR INDICES

Now, what’s the problem this quantization solves?

We’re trying to solve the RAM problem (For NOW)!



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Let's Connect the Dots!

- You have 100 million vectors
- each with 1536 dimensions

What is the size per Vector?

100 Million Vectors, 1536 Dimensions

Memory Usage by Data Type

Note: Generated Image for cleaner representation

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Now, it makes everything Possible !
And, Not Expensive !

Let’s Play a Wordle GAME!

Documentation References

Quantization (BBQ, Scalar Quantization)

- https://www.elastic.co/search-labs/blog/better-binary-quantization-lucene-elasticsearch
- https://www.elastic.co/search-labs/blog/optimized-scalar-quantization-elasticsearch
- https://www.elastic.co/search-labs/blog/bit-vectors-elasticsearch-bbq-vs-pq
- https://www.elastic.co/docs/reference/elasticsearch/mapping-reference/bbq
- https://www.elastic.co/docs/solutions/search/vector/knn

DiskBBQ

- https://www.elastic.co/search-labs/blog/diskbbq-elasticsearch-introduction
- https://www.elastic.co/search-labs/blog/elasticsearch-latency-low-memory-diskbbq-hnswbbq-benchmark

HNSW Graphs

- https://www.elastic.co/search-labs/blog/hnsw-graph
- https://www.elastic.co/search-labs/blog/hnsw-graphs-speed-up-merging

Related Benchmarks/Comparisons

- https://www.elastic.co/search-labs/blog/elasticsearch-bbq-vs-opensearch-fais

Let’s Connect Again!